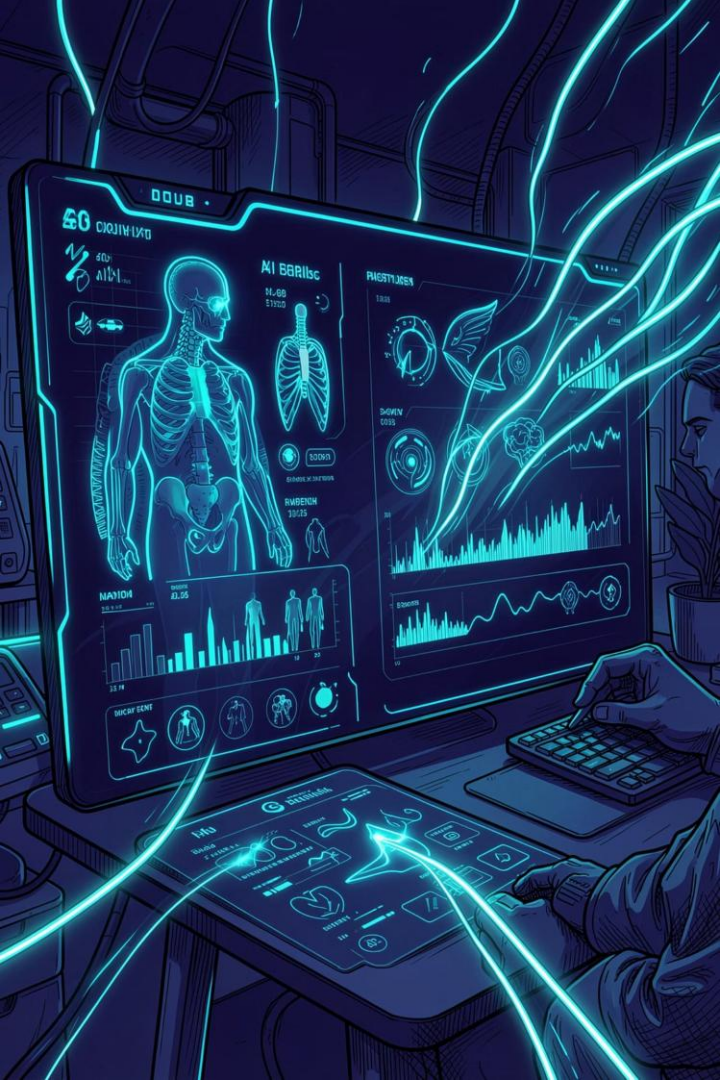




PRODUCT LAUNCH · 2026

AI Health Companion 1.0

Hybrid AI-Based Preventive Healthcare & Wellness Assistance System



THE CHALLENGE

Healthcare Is Reactive. It Should Be Predictive.

Late Detection

Most conditions are diagnosed at advanced stages — when treatment is costly and outcomes are uncertain.

Access Gap

Millions lack consistent access to preventive care, specialists, or timely medical guidance.

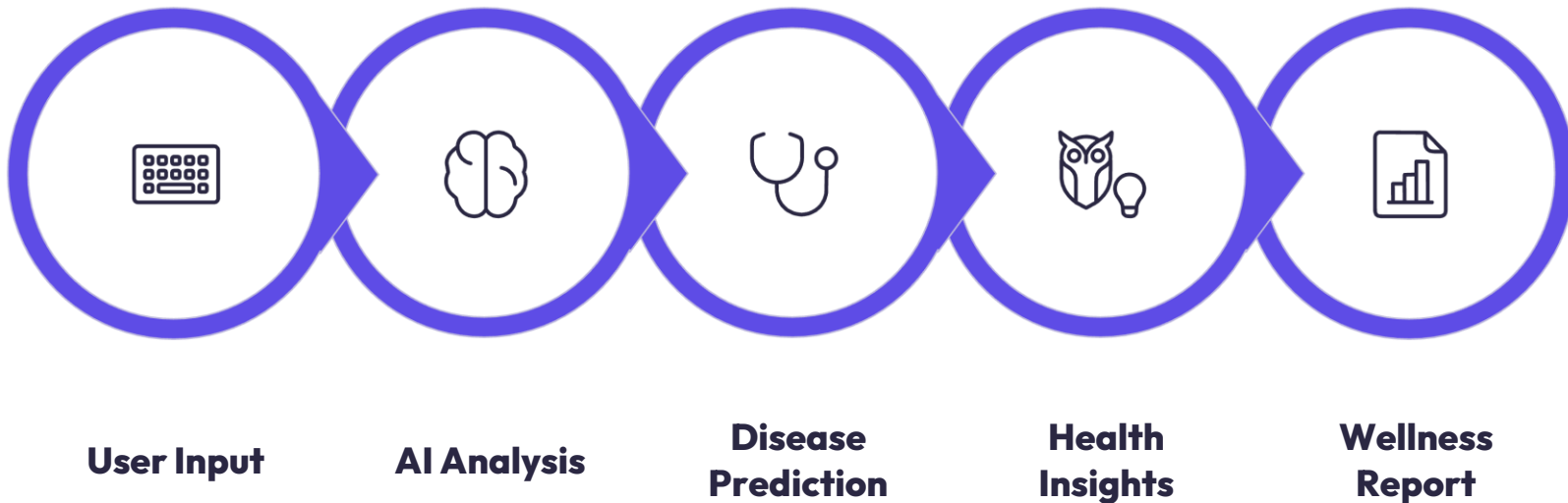
Data Overload

Clinicians and patients are overwhelmed by fragmented data with no intelligent synthesis layer.

SOLUTION

One Platform. Complete Preventive Intelligence.

AI Health Companion transforms raw health data into actionable, personalized wellness intelligence — before symptoms appear.

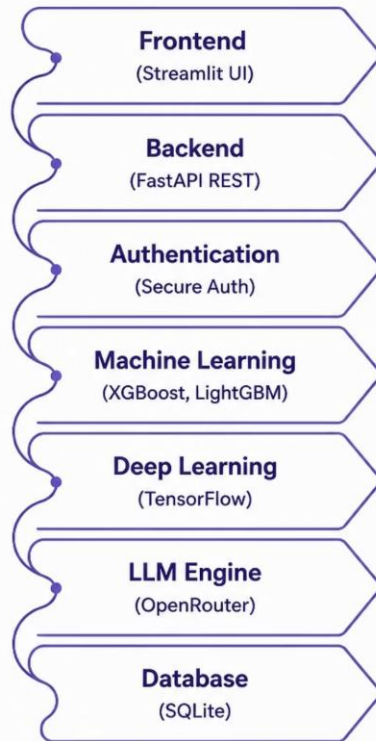


From first input to final report — every step is powered by a hybrid AI pipeline combining classical ML, deep learning, and large language models.

System Architecture

A clean, modular architecture built for scalability — each layer operates independently while contributing to a unified intelligence engine.

- **Frontend** — Streamlit-powered responsive UI
- **Backend** — FastAPI REST API with async support
- **Auth** — Secure session-based authentication
- **ML / DL** — Hybrid prediction ensemble
- **LLM** — OpenRouter for natural language insights
- **Database** — SQLite for structured health records



Built With Best-in-Class Tools



Streamlit

Interactive frontend & dashboards



TensorFlow

Deep learning model training & inference



LightGBM

Lightweight gradient boosting



Plotly

Interactive data visualizations



FastAPI

High-performance async REST API



XGBoost

Gradient boosting for tabular prediction



SQLite

Lightweight embedded database



OpenRouter

LLM integration for health insights

Core Features



Authentication

Secure, session-based login with protected health records



Hybrid Disease Prediction

ML + Deep Learning ensemble for multi-condition risk scoring



AI Health Assistant

LLM-powered conversational guidance tailored to your profile



Interactive Dashboard

Real-time charts, risk visualizations, and trend tracking



PDF Reports

Exportable, shareable wellness reports with AI-generated summaries



History Management

Full audit trail of past predictions, inputs, and reports

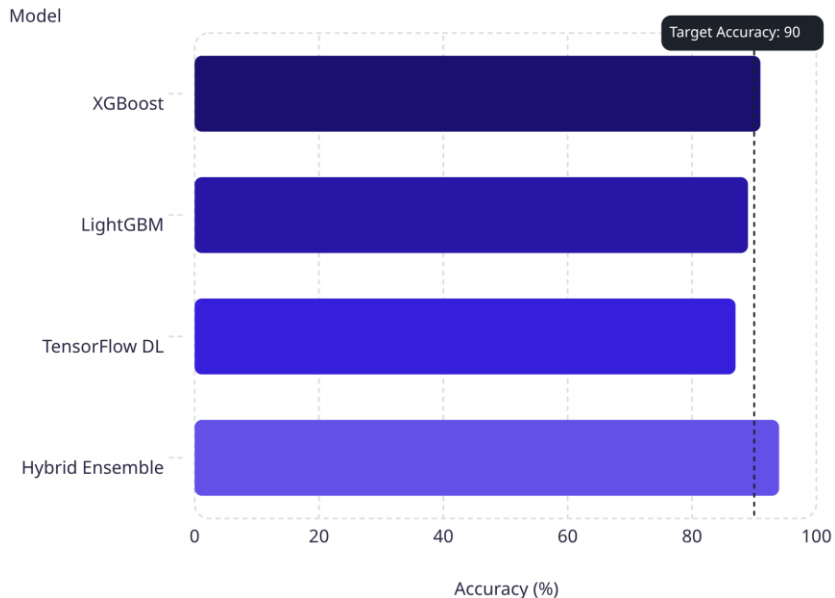
The Hybrid AI Engine

Two parallel intelligence streams — classical ML and deep learning — converge into a unified risk score, then interpreted by a large language model for human-readable insights.

- The hybrid approach outperforms single-model baselines by combining the interpretability of gradient boosting with the pattern depth of neural networks.



Results & Performance



What the Numbers Mean

The hybrid ensemble consistently outperforms individual models — demonstrating the power of combining gradient boosting with deep learning for medical prediction tasks.

94% Accuracy

Hybrid ensemble on test data

High Confidence

Calibrated probability scores per prediction

Real-World Applications



Hospitals & Clinics

Triage support and early risk flagging for clinical teams



Preventive Care

Population-level screening and proactive health monitoring



Medical Research

Scalable data pipelines for epidemiological studies



Medical Education

Training tool for students and healthcare professionals



Future Expansion

Multi-modal inputs, wearables integration, and clinical APIs

THANK YOU

Built for the Future of Preventive Health.

AI Health Companion 1.0 is a demonstration of what hybrid AI can achieve when engineering rigor meets real-world healthcare challenges.

Portfolio

Full project case study & demo

GitHub

Source code & documentation

LinkedIn

Connect & collaborate

